

A Shannon–Haar Counterexample to Interpolation-Unitary Normalization

Candidate result for arXiv:math/0604619, Problem 4

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Status. Candidate counterexample, likely valid, pending expert review. The result gives a negative answer to Problem 4 of Larson’s source paper.

1 Source problem

Let

$$(Df)(x) = \sqrt{2} f(2x), \quad (Tf)(x) = f(x - 1)$$

on $L^2(\mathbb{R})$. For dyadic orthonormal wavelets ψ, η , the interpolation unitary V_ψ^η is the unitary determined by

$$V_\psi^\eta D^j T^k \psi = D^j T^k \eta \quad (j, k \in \mathbb{Z}).$$

Larson asks whether V_ψ^η must normalize $\{D, T\}'$ [1, Problem 4, p. 28].

is necessarily an **MSF** wavelet. (In other words: is a minimal support set in the frequency domain necessarily a wavelet set).

Problem 4: [Normalization] (See section 3.3.1.) If $\{\psi, \eta\}$ is a pair of dyadic orthonormal wavelets, does the interpolation unitary V_ψ^η normalize $\{D, T\}'$? As mentioned above, the answer is yes if ψ and η are s -elementary wavelets. This problem makes sense for orthonormal wavelets in higher dimensions for matrix dilation factors, and for other scalar dilations in one dimension. We know of no counterexample for any of these cases. However, the problem might just lie in the fact that in most cases, other than for wavelet sets, we have no reasonable techniques for doing the computations. This problem was also discussed in context in Section 3.3.1, and it could be the most important problem remaining in the direction of further development of the unitary system approach to wavelet theory.

Figure 1: Problem 4 in the source PDF, page 28.

The source proves the Fourier-side commutant characterization

$$\mathcal{F}\{D, T\}'\mathcal{F}^{-1} = \{M_h : h \in L^\infty(\mathbb{R}), h(s) = h(2s) \text{ a.e.}\} \quad (1)$$

[1, Theorem 4.1, pp. 12–13].

2 Definitions and the proposed family

Use the unitary Fourier transform

$$\widehat{f}(s) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-isx} f(x) dx.$$

The Shannon wavelet ψ_S is defined by

$$\widehat{\psi}_S = \frac{1}{\sqrt{2\pi}} \mathbf{1}_E, \quad E = [-2\pi, -\pi) \cup [\pi, 2\pi).$$

The argument applies to every dyadic orthonormal wavelet η such that $\widehat{\eta} \neq 0$ almost everywhere. For an explicit example, take the standard Haar wavelet

$$\psi_H = \mathbf{1}_{[0,1/2)} - \mathbf{1}_{[1/2,1)}.$$

Both are dyadic orthonormal wavelets. A direct integration gives

$$\widehat{\psi}_H(s) = \frac{(1 - e^{-is/2})^2}{is\sqrt{2\pi}}, \quad (2)$$

with the value at $s = 0$ defined by continuity. In particular, $\widehat{\psi}_H$ is nonzero almost everywhere: its zeros form a countable set.

Let p be the 2π -periodic half-torus mask

$$p(s) = \mathbf{1}_{[\pi, 2\pi)}(s \bmod 2\pi). \quad (3)$$

Then p is not 2-dilation-periodic. Indeed, for $s \in (\pi/2, \pi)$ one has $p(s) = 0$ and $p(2s) = 1$.

3 Idea of the proof

The positive-frequency projection is a member of the commutant because its symbol is invariant under $s \mapsto 2s$. On the two Shannon frequency bands, that projection is exactly the periodic mask p . Periodicity means that p acts through a translate-coefficient Fourier series. The interpolation unitary preserves those coefficients when it replaces every Shannon translate by the corresponding target-wavelet translate, so the conjugated projection sends the target mother wavelet to $p\widehat{\eta}$.

If the interpolation unitary normalized the commutant, this conjugated projection would be M_h for a dilation-periodic h . Its action on the target mother wavelet would force $h = p$ almost everywhere because $\widehat{\eta} \neq 0$ almost everywhere. This is impossible because p is not dilation-periodic.

4 Formal result and proof

Theorem 1. *Let η be any dyadic orthonormal wavelet satisfying $\widehat{\eta} \neq 0$ almost everywhere, and let $V = V_{\psi_S}^\eta$ be the interpolation unitary from the Shannon wavelet to η . Then V does not normalize $\{D, T\}'$. In particular, this holds for $\eta = \psi_H$, so Problem 4 in [1] has a negative answer.*

Proof. Work in the Fourier domain and write

$$\widehat{V} = \mathcal{F}V\mathcal{F}^{-1}, \quad P_+ = M_q, \quad q = \mathbf{1}_{(0,\infty)}.$$

Since $q(2s) = q(s)$ almost everywhere, (1) gives $P_+ \in \mathcal{F}\{D, T\}'\mathcal{F}^{-1}$.

On the Shannon support E , the functions q and p agree. Hence

$$P_+ \widehat{\psi}_S = M_p \widehat{\psi}_S. \quad (4)$$

Let $(c_k)_{k \in \mathbb{Z}}$ be the Fourier coefficients of p , with signs chosen so that

$$p(s) = \sum_{k \in \mathbb{Z}} c_k e^{-iks} \quad \text{in } L^2([0, 2\pi)).$$

The fixed-scale translates of either wavelet are orthonormal. Therefore their synthesis maps are isometries from $\ell^2(\mathbb{Z})$ onto their respective closed translate spans. Applying the finite Fourier sums first and then taking the L^2 limit gives

$$M_p \widehat{\psi}_S = \sum_{k \in \mathbb{Z}} c_k \widehat{T}^k \widehat{\psi}_S, \quad (5)$$

$$M_p \widehat{\eta} = \sum_{k \in \mathbb{Z}} c_k \widehat{T}^k \widehat{\eta}. \quad (6)$$

By the defining property of the interpolation unitary, $\widehat{V} \widehat{T}^k \widehat{\psi}_S = \widehat{T}^k \widehat{\eta}$. Since $\widehat{V}$ is unitary, (4)–(6) imply

$$\widehat{V} P_+ \widehat{\psi}_S = M_p \widehat{\eta}. \quad (7)$$

Set $R = \widehat{V} P_+ \widehat{V}^*$. Since $\widehat{V}^* \widehat{\eta} = \widehat{\psi}_S$, equation (7) says

$$R \widehat{\eta} = M_p \widehat{\eta}. \quad (8)$$

Suppose, toward a contradiction, that V normalizes $\{D, T\}'$. Then R belongs to the Fourier-side commutant. By (1), $R = M_h$ for some $h \in L^\infty(\mathbb{R})$ satisfying $h(s) = h(2s)$ almost everywhere. Equation (8) yields

$$(h - p) \widehat{\eta} = 0 \quad \text{a.e.}$$

The hypothesis $\widehat{\eta} \neq 0$ almost everywhere gives $h = p$ almost everywhere. This contradicts the failure of $p(s) = p(2s)$ on $(\pi/2, \pi)$. Therefore V does not normalize the commutant. Formula (2) verifies the hypothesis for the explicit choice $\eta = \psi_H$. $\square$

5 Verification notes

The proof was attacked at the following convention-sensitive points.

- If the opposite orientation for V_ψ^η is adopted, the argument uses the inverse conjugation. A unitary normalizes an algebra if and only if its adjoint does, so the conclusion is unchanged.
- A Fourier-series sign reversal only reindexes the translate coefficients. The mask p is real, and the transported multiplier identity is unchanged.
- Values at the endpoints of the half-torus intervals and at the countable zero set of $\widehat{\psi}_H$ are null-set issues.
- No numerical evidence is used; the counterexample is exact.

The independent step-by-step verifier report is included as `verification.md`. Current verdict: **likely valid**.

6 Scope and novelty check

This counterexample addresses exactly the one-dimensional dyadic question. It does not make claims about other scalar dilations or higher-dimensional matrix dilations.

A bounded novelty search on 27 August 2026 covered the run’s lightweight registry, solution, attempt, and proof-gap indexes, together with exact and close arXiv/web searches for “interpolation unitary”, “normalize commutant”, “Shannon Haar”, and Larson’s Problem 4. The search found the

source paper and the related 2006 exposition arXiv:math/0604615, but no later explicit solution or counterexample. This was not an exhaustive citation-database search; novelty is therefore provisional.

7 Human-review recommendation

Send to a wavelet/operator-algebra expert. The highest-value review step is to check equations (5)–(7), namely that the same periodic Fourier coefficients synthesize the masked Shannon and target mother wavelets and are transported term by term by the interpolation unitary.

References

- [1] D. R. Larson, *Unitary systems and wavelet sets*, arXiv:math/0604619 (2006).